Lab U3 - The `urdf_tutorial` robots (R2-D2 style, eight steps)

Aim. Build the official `urdf_tutorial` package from its repository and look at its eight example robots, from one cylinder to a macro-generated robot, to see how each URDF/xacro feature changes what RViz2 draws.

Time. 45 min. **Needs.** Only this folder. (The package is pure Python/XML: no compiler.) The repository is downloaded by `pixi run build-urdf`. Source: `upstream\urdf_tutorial-ros2` (branch `ros2`), built into `upstream_ws\urdf`.

This folder is a complete project

Everything this lab needs is in this folder. Nothing from the other labs is required.

Requirements: Windows 10/11, PowerShell, pixi (`winget install --id prefix-dev.pixi -e --source winget`), and internet access once, to download `ros/urdf_tutorial` from GitHub.

In this folder	
`upstream\`	the original repository (downloaded by the first build task, pinned commit); `upstream_ws\` its build
`scripts\`	helper scripts and the automated tests
`pixi.toml`	the environment and every `pixi run ...` command; `README.md` this manual; `tools\replay.ps1` re-runs the manual's commands as a self-test

Set it up once. `pixi install` puts ROS 2 Jazzy into `C:\rlabs` (a few GB; pixi's package cache makes the second lab on the same machine fast). Open PowerShell in this folder and run:

```
pixi install
pixi run verify
```

Run one lab at a time, and everything from this folder.

Theory

File	New feature
`01-myfirst.urdf`	one link with a cylinder
`02-multipleshapes.urdf`	two links, one **fixed joint**
`03-origins.urdf`	joint **origin** (`xyz`, `rpy`): where the child sits
`04-materials.urdf`	**materials** (colours)
`05-visual.urdf`	different shapes, **meshes** for the head
`06-flexible.urdf`	**movable joints**: continuous, revolute, prismatic - sliders appear
`07-physics.urdf`	`collision` and `inertial` tags (needed by a physics simulator)
`08-macroed.urdf.xacro`	**xacro**: constants, macros, math - same robot, much shorter

`display.launch.py` starts `robot_state_publisher` (URDF -> TF), `joint_state_publisher_gui` (sliders for the movable joints) and RViz2. The robot is chosen with the launch argument `model:=urdf/<file>`.

Procedure

Shortcut for commands that need this package (re-type in each new terminal):

```
function urdf { pixi run cmd /c scripts\in_upstream.bat urdf @args }
```

1. Build.

```
pixi run build-urdf
```

2. T1 - the first robot (a single blue cylinder; fixed frame `base_link`).

```
pixi run demo-urdf
```

3. T2 - what was published.

```
pixi run ros node list --no-daemon --spin-time 5
pixi run ros topic list --no-daemon --spin-time 5
```

4. T1 - the movable robot. Stop T1 (Ctrl+C), then start the sixth robot. Move the sliders: the head turns, the gripper opens, the arm slides.

```
# Ctrl+C in T1
```

```
pixi run demo-urdf model:=urdf/06-flexible.urdf
```

```
pixi run ros topic echo /joint_states --no-daemon --once --spin-time 5
```

(Only the movable joints are in `/joint_states`; the fixed ones never change.)

5. T1 - the macro version. Compare 07 and 08: same robot, one file written with xacro.

```
# Ctrl+C in T1
```

```
pixi run demo-urdf model:=urdf/08-macroed.urdf.xacro
```

```
pixi run ros topic echo /robot_description --no-daemon --once --spin-time 5 --field data
```

Count the lines of the two source files, and expand the xacro to plain URDF to see what the macros produce:

```
(Get-Content upstream\urdf_tutorial-ros2\urdf\07-physics.urdf).Count    # 417
(Get-Content upstream\urdf_tutorial-ros2\urdf\08-macroed.urdf.xacro).Count    # 237
pixi run xacro upstream\urdf_tutorial-ros2\urdf\08-macroed.urdf.xacro -o r2d2.urdf
pixi run check_urdf r2d2.urdf
Remove-Item r2d2.urdf
```

```
# Ctrl+C in T1
```

6. Try the others. Run `pixi run demo-urdf model:=urdf/0N-....urdf` for 01-05 and 07. Before each, predict what RViz2 will draw.

Observations (recorded)

Item	Value
Nodes with `01-myfirst`	`/robot_state_publisher`, `/rviz`; topics `/robot_description`, `/tf_static` (no `/joint_states`: the robot has no movable joint). With `06-flexible` the slider node `/joint_state_publisher` also appears and `/joint_states` is published
`06-flexible` joints in `/joint_states`	`head_swivel`, `gripper_extension`, `left_gripper_joint`, `right_gripper_joint` and the four wheel joints
Source size	`07-physics.urdf` 417 lines, `08-macroed.urdf.xacro` 237 lines (the robot is named `macroed`); the xacro expands to a URDF with root link `base_link`

Verification

```
pixi run test-urdf
```

17 checks: for each of the eight robots the package's own `display.launch.py` is started; the published `/robot_description` equals the source file (after xacro: same link and joint counts) and every link resolves in TF.

Questions

1. Which joint types appear in `06-flexible.urdf` and which of them gets a slider?
2. In `03-origins.urdf`, what is the difference between the joint `origin` and the visual `origin`?
3. How many lines does the macro version save (417 vs 237), and which repeated block does the macro replace?
4. Why does `07-physics.urdf` need `inertial` and `collision` tags when RViz2 does not?
5. *(Extension)* Add a camera link to `08-macroed.urdf.xacro` using the same macro style (edit a *copy*, not the clone).

If it does not work

Symptom	Fix
`malformed launch argument 'model:'`	you are using an older `scripts\in_upstream.bat`; update it (it must read `%*`)
`package 'urdf_tutorial' not found`	`pixi run build-urdf`
RViz2 grey	set *Fixed Frame* to `base_link`